Week 2. By Deng Yufan.

Prob 1. Find λ terms representing logical or, not.

or: $\lambda xy.\, xty$.

not: $\lambda x.\, xft$.

Prob 2. Prove.

a. $\text{add}\ \bar{m}\ \bar{n} \twoheadrightarrow_\beta \overline{m+n}$.

$$
\begin{aligned}
\text{add}\ \bar{m}\ \bar{n} &\equiv (\lambda mnfx.\, mf(nfx))\ \bar{m}\ \bar{n} \\
&\cancel{\to_\beta (\lambda nfx.\, \bar{m} f(nfx))\, n} \\
&\to_\beta \lambda fx.\, \bar{m} f(\bar{n} f x) \\
&\equiv \lambda fx.\, \bar{m} f((\lambda fx.\, f^n x) f x) \\
&\to_\beta \lambda fx.\, \bar{m} f(f^n x). \\
&\equiv \lambda fx.\, (\lambda fx.\, f^m x) f (f^n x) \\
&\equiv \lambda fx.\, f^{m+n} x. \\
&\equiv \overline{m+n}.
\end{aligned}
$$

b. $\text{mult}\ \bar{m}\ \bar{n} \twoheadrightarrow_\beta \overline{m \cdot n}$.

$$
\begin{aligned}
\text{mult}\ \bar{m}\ \bar{n} &\equiv (\lambda mnf.\, m(nf))\ \bar{m}\ \bar{n} \\
&\to_\beta \lambda f.\, \bar{m}(\bar{n} f). \\
&\equiv \lambda f.\, \bar{m}((\lambda fx.\, f^n x) f) \\
&\to_\beta \lambda f.\, \bar{m}(\lambda x.\, f^n x) \\
&\equiv \lambda f.\, (\lambda fx.\, f^m x)(\lambda x.\, f^n x) \\
&\to_\beta \lambda f.\, (\lambda x.\, (\lambda x'.\, f^n x')^m x) \\
&\equiv \lambda fx.\, f^{m \cdot n} x. \\
&\equiv \overline{m \cdot n}.
\end{aligned}
$$

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Prob 3. Compute following terms. Are they strongly normalizable?

a. $(\lambda xy.yx)\big((\lambda x.xx)(\lambda x.xx)\big)(\lambda xy.y)$

$\to_\beta (\lambda xy.y)\big((\lambda x.xx)(\lambda x.xx)\big)$

$\to_\beta \lambda y.y$.

is not strongly normalizable since $(\lambda x.xx)(\lambda x.xx) \to_\beta (\lambda x.xx)(\lambda x.xx) \to_\beta \cdots$.

b. $(\lambda xy.yx)(kk)(\lambda x.xx)$ $\quad [k = \lambda xy.x]$

$\to_\beta (\lambda x.xx)(kk)$

$\to_\beta (kk)(kk)$

$\equiv \big((\lambda xy.x)k\big)\big((\lambda xy.x)k\big)$

$\to_\beta (\lambda y.k)(\lambda y.k)$

$\to_\beta k$.

Prob 4. Find a representation of following func.

a. $f(n) = \begin{cases} \text{true} & n \text{ is even} \\ \text{false} & n \text{ is odd.} \end{cases}$

$\lambda n.n \text{ not } t$. or say.

$\lambda n.n(\lambda x.xft)t$.

b. $\exp(n,m) = n^m$.

$\lambda nm.m(\text{mul } \bar{n})\bar{1}$ or say, By LLM. we find.

$\lambda nm.m(\text{mul } n)(\lambda fx.fx)$. $\lambda nm.mn$.

(*) c. $\text{pred}(n) = \begin{cases} 0 & n=0 \\ n-1 & \text{otherwise} \end{cases}$. By LLM. we find.

$\lambda n.n\big(\lambda xy.y(\text{succ}(xt))(xt)\big)(\lambda x.x\bar{0}\bar{0})f$. $\lambda nfx.n(\lambda gh.h(gf))(\lambda u.x)(\lambda u.u)$

I did it by myself!

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Prob 5. $\to_1, \to_2$ commute: $s \to_1 t_1$. $s \to_2 t_2$. $\overset{\exists t.}{\Rightarrow}$ $t_1 \to_2 t$. $t_2 \to_1 t$.

$\to_{12} = \to_1 \cup \to_2$.

$\to_1$. $\to_2$ has Church-Rosser Prop.

Prove: $\to_{12}$ has C-R Prop.

Lemma. $s \twoheadrightarrow_1 t_1$. $s \twoheadrightarrow_2 t_2$ $\overset{\exists t.}{\Rightarrow}$ $t_1 \twoheadrightarrow_2 t$. $t_2 \twoheadrightarrow_1 t$.
furthermore, steps are the same, respectively.

Let steps of $s \twoheadrightarrow_1 t_1$ be n_1.
$s \twoheadrightarrow_2 t_2$ be n_2.

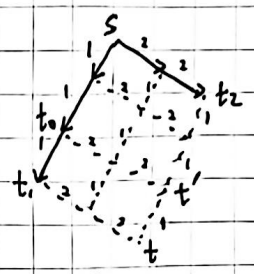

fig 1.

Induction ~~[illegible]~~ n_1.

As basis: when $n_1 = 0$, $s \twoheadrightarrow_1 s$. $s \twoheadrightarrow_2 t_2$,

then let $t = t_2$. $s \twoheadrightarrow_2 t_2$, $t_2 \twoheadrightarrow_1 t_2$.

As induction: assume n_1 and prove n_1+1.

$s \twoheadrightarrow_1 t_0 \to_1 t_1$. $s \twoheadrightarrow_2 t_2$. ($s \twoheadrightarrow t_1$ take n_1+1 step. $s \twoheadrightarrow t_0$ take n_1 step)

then $\overset{\exists t'.}{}$ $t_0 \twoheadrightarrow_2 t'$. $t_2 \twoheadrightarrow_1 t'$.

Induction n_2. when fix $n_1 = 1$.

As basis: when $n_2 = 0$. obvious.

As induction: assume n_2 and prove n_2+1.

$s \to_1 t_1$. $s \twoheadrightarrow_2 t_0 \to_2 t_2$.

then $\exists t'$ $t_1 \twoheadrightarrow_2 t'$. $t_0 \overset{\to}{\to}_1 t'$.

since $t_0 \to_1 t'$. $t_0 \to_2 t_2$.

then $\exists t$. $t' \to_2 t$. $t_2 \to_1 t$.

thus $t_1 \twoheadrightarrow_2 t' \to_2 t$. $t_2 \to_1 t$.

since $t_0 \to_1 t_1$. $t_0 \twoheadrightarrow_2 t'$. then $\exists t$. $t_1 \twoheadrightarrow_2 t$. $t' \to_1 t$.
thus $t_1 \twoheadrightarrow_2 t$. $t_2 \twoheadrightarrow_1 t' \to_1 t$.

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Formally proving is kind of annoying. We briefly write the second part with figure.

Thm. $s \twoheadrightarrow_{12} t_1$. $s \twoheadrightarrow_{12} t_2$. $\Rightarrow \exists t.\ t_1 \twoheadrightarrow_{12} t.\ \ t_2 \twoheadrightarrow_{12} t$.

Similarly induct twice. but we divide steps into component of continuous steps of 1 and 2 instead of single step.

fig 2.

Since $\twoheadrightarrow_1$ & $\twoheadrightarrow_1$, $\twoheadrightarrow_2$ & $\twoheadrightarrow_2$ is given in condition.

$\twoheadrightarrow_1$ & $\twoheadrightarrow_2$ is proven in lemma.

The Thm can be proven easily.

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